



Indian Association for the Cultivation of Science
(Deemed to be university under the *de novo* category)

Integrated BSMS Program

Mid-Semester Examination-2025 (Autumn Semester 2-BSMS)

Subject: Quantum and Thermal Physics
Full Marks: 25

Subject Code(s): PHS 2101
Time Allotted: 2hr

Answer any five questions from the seven.

1. In a classical physics formulation of an atom, if a negatively charged electron revolves around a positively charged nucleus in a circular orbit, can the orbit be stable ? Why ? Explain briefly but precisely the motion of the electron in this situation. If now you consider a bunch of such classical atoms in a box, can this system be in thermal equilibrium ? Why ? What must you do to ensure that the system can reach thermal equilibrium eventually ? [1+1+1+2]
2. While motivating the light-quantum hypothesis, Einstein, in his seminal 1905 paper, speaks about 'contradictions' that arise in the classical formulation of blackbody radiation, between the continuum-field based Maxwell theory of EM and the observed quantization of electric charge. What contradictions do you think Einstein is referring to, from your knowledge of classical blackbody radiation ? If you were to open the viewing window of a steel furnace in a steel plant in operation, how would these contradictions appear to you ? Please answer briefly but precisely. [2+3]
3. S N Bose, in his pioneering 1924 paper on derivation of the Planck law of blackbody radiation, is critical of Einstein's attempts to derive the Planck law on the basis of his light-quantum hypothesis. What, according to Bose, is wrong with Einstein's derivation ? Briefly sketch how Bose attempts to get around these difficulties. How well does Bose succeed in your opinion ? [2+2+1]
4. To address the key issue of placing N_γ photons (with frequency ω) in N_c cells in phase space, Bose derives the formula for the number of arrangements to be

$$\Omega_\gamma = \frac{N_c!}{n_0!n_1!\dots}$$

where n_0 is the number of cells with zero photons in them, n_1 is the number of cells with one photon in each, and so on. Justify this formula, both mathematically and physically, as best as you can. The mathematical problem that Bose then solves is to *extremize* this number of arrangements, subject to two physical constraints. What constraints are these? The result of this mathematical exercise is to obtain the extremal value of the number of cells with r photons in each of them, given by

$$n_r = \bar{n} \exp(-\beta \hbar \omega)$$

where $\bar{n}$ is a constant, independent of the frequency ω . How does one derive $\bar{n}$ and what is its value in terms of N_c ? [1+2+2]

5. Consider a diatomic molecule of atoms of masses m_A and m_B . Show that the average kinetic energy, of the centre-of-mass motion, of a gas of such diatomic molecules at equilibrium at a temperature T , is $(3/2)kT$ where k is the Boltzmann constant. Mention how the the assumption of vanishing of the average of the scalar product of the CM velocity and the relative velocity vectors plays a role in this derivation. [4+1]
6. A deflated bicycle tyre has air ($\gamma = 1.4$) at atmospheric pressure (about 10^5 N/cm^2) at room temperature (27 degrees celsius). If you inflated the tyre to a pressure of about 3.5 atmospheres, what would be the temperature in degrees celsius if that pressurised air from the same tyre tube is suddenly released? [5]
7. Imagine a tall vertical column of a gas whose density $\rho = \rho(h)$, i.e., it is function of the height of the observation point. Show that the pressure as a function of height follows the differential equation

$$\frac{dP}{dh} = -\rho(h)g$$

where g is the acceleration due to gravity. Assume that the entire column is in equilibrium at a temperature T . Solve the equation for a gas of molecular weight M . [3+2]